

Find the x and y intercepts from a graph. (See the midterm review for examples)

1. Find all the times the graph intersects the x -axis. These are your x -intercepts.
2. Find all the times the graph intersects the y -axis. These are your y -intercepts.

Solve an equation with fractions. ex. $\frac{2x}{6} + 5 = \frac{3x}{9}$

1. Multiply by the Lowest Common Denominator (LCD). In this case, 18. Note BOTH terms on the left got multiplied by 18.

$$\begin{aligned} 18\left(\frac{2x}{6} + 2\right) &= \left(\frac{x}{9}\right) 18 \\ \frac{36x}{6} + 36 &= \frac{18x}{9} \\ 6x + 36 &= 2x \end{aligned}$$

2. Now just solve.

$$\begin{aligned} 4x &= -36 \\ x &= -\frac{36}{4} = -9 \end{aligned}$$

Solve a formula for a specified variable. ex. $P = 2L + 2W$ for L .

1. Just solve like you do for an x , and treat the other letters like numbers.

$$\begin{aligned} P &= 2L + 2W \\ P - 2W &= 2L \\ \frac{P - 2W}{2} &= L \end{aligned}$$

2. Note: The 2 divides BOTH factors, and the you DO NOT cancel anything in $2W$.

Solve using the square root property. ex. $(x - 4)^2 + 1 = 2$.

1. Get the squared part by itself:

$$(x - 4)^2 = 1$$

2. Use a square root; Don't forget the $\pm$.

$$\begin{aligned} \sqrt{(x - 4)^2} &= \pm\sqrt{1} \\ x - 4 &= \pm 1 \end{aligned}$$

3. Solve.

$$\begin{aligned} x &= \pm 1 + 4 \\ x = 5 \quad \text{and} \quad x = 3 \end{aligned}$$

Solve a radical. ex. $\sqrt{x - 1} + 2 = 5$.

1. Get the radical by itself,

$$\begin{aligned} \sqrt{x - 1} + 2 &= 5 \\ \sqrt{x - 1} &= 3 \end{aligned}$$

2. Use a square root, simplify, and solve.

$$\begin{aligned} (\sqrt{x - 1})^2 &= 3^2 \\ x - 1 &= 9 \\ x &= 10 \end{aligned}$$

3. Check! $\sqrt{10 - 1} = 9\checkmark$

Solve the absolute value or indicate that it has no solution. ex. $|x + 2| - 2 = 10$

1. Get the absolute value by itself.

$$\begin{aligned} |x + 2| - 2 &= 10 \\ |x + 2| &= 12 \end{aligned}$$

2. Solve for + and -.

$$\begin{aligned} x + 2 &= -12 & \text{and} & & x + 2 &= 12 \\ x &= -14 & & & x &= 10 \end{aligned}$$

3. Check! $|-14 + 2| - 2 = 10\checkmark$ and $|10 + 2| - 2 = 10\checkmark$

Solve by factoring. ex. $3x^2 = -3x + 6$

1. Set the equation equal to 0.

$$3x^2 + x - 2 = 0$$

2. Factor any common multiples.

$$3(x^2 + x - 2) = 0$$

3. Factor and solve.

$$\begin{aligned} 3(x + 2)(x - 1) &= 0 \\ x &= -2 & \text{and} & & x &= 1 \end{aligned}$$

Solve the equation using the quadratic formula. ex. $3x^2 = -2x + 2$.

1. Set the equation equal to 0.

$$3x^2 + 2x - 2 = 0$$

2. Identify the a, b, c in $ax^2 + bx + c = 0$

$$a = 3, b = 2, c = -2$$

3. Use the quadratic formula, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, and simplify.

$$x = \frac{-2 \pm \sqrt{(2)^2 - 4(3)(-2)}}{2(3)} = \frac{-2 \pm \sqrt{28}}{6} = \frac{-1 \pm \sqrt{7}}{3}$$

Solve the polynomial by factoring and using the zero principle. ex. $x^3 = -x^2 + 2x$.

1. Set the equation equal to 0.

$$x^3 + x^2 - 2x = 0$$

2. Factor out as many x 's as you can.

$$x(x^2 + x - 2) = 0$$

3. Factor.

$$x(x + 2)(x - 1) = 0$$

4. Solve. DO NOT forget the $x = 0$ part!

$$x = 0, x = -2, x = 1$$

Use the 5 step strategy for solving the word problem. When 3 times the number is subtracted 7 times the number, the result is 44.

1. Summarize: "3 times a number" means, $3x$. "7 times a number", means $7x$. "Subtracted from" means, $7x - 3x$. "Result is 44" means, $= 44$.
2. Write an equation: $7x - 3x = 44$
3. Solve: $x = 11$.
4. Check: $7(11) - 3(11) = 44\checkmark$
5. State your answer as a sentence (this isn't applicable here).

Solve the word problem. ex. A car rental place charges \$300 a week to rent a car plus an additional \$0.50 per mile. How many miles can you drive in a week for \$400.

1. Write as an equation.

$$300 + 0.50x = 400$$

2. Solve.

$$\begin{aligned} 0.50x &= 100 \\ x &= \frac{100}{0.50} = \$200 \end{aligned}$$

Find the Domain and the Range.

1. Domain= $x's$; Range= $y's$.
2. From a table or list, just find the $x's$ and $y's$. ex.

$$\begin{aligned} &\{(1, 2), (0, 3), (-1, 4), (-2, 5)\} \\ &\text{domain} = \{1, 0, -1, -2\} \\ &\text{range} = \{2, 3, 4, 5\} \end{aligned}$$

3. From a graph, look at the horizontal values for the domain ($x's$) and the vertical values for the range ($y's$). If they appear to go off the graph, then use $\pm\infty$ (See the midterm review for examples).

Determine if the relation is a function. ex. $\{(1, -2), (2, -1), (2, 0), (3, 1)\}$

1. Check that each x , in (x, y) , is paired with only one y value.

$$\begin{aligned} 1 &\rightarrow -2 \\ 2 &\rightarrow -1 \\ 2 &\rightarrow 0 \\ 3 &\rightarrow 1 \end{aligned}$$

2. If any x is paired with two different $y's$ then the relation is NOT a function.

$$\begin{aligned} 2 &\rightarrow -1 \\ 2 &\rightarrow 0 \end{aligned}$$

So this is NOT a function.

3. Be suspicious if an x appears more than once.

Find the slope passing through the given points. ex, $(1, 3)$ and $(0, 5)$

1. Use the formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Identify (x_1, y_1) and (x_2, y_2) .

$$x_1 = 1, y_1 = 3 \text{ and } x_2 = 0, y_2 = 5$$

2. Plug the values into the formula.

$$m = \frac{5 - 3}{0 - 1} = \frac{2}{-1} = -2$$

Use the information to write an equation for the line in point-slope formula. ex. Slope of -2 , passing through $(1, 3)$.

1. Use the formula, $y - y_1 = m(x - x_1)$. Identify m , x_1 , and y_1 .

The slope is $m = -2$, and the point gives $x_1 = 1$ and $y_1 = 3$.

2. Plug the values into the formula.

$$y - 3 = -2(x - 1)$$

3. Simplify and write as $y = mx + b$.

$$\begin{aligned} y &= -2x + 2 + 3 \\ y &= -2x + 5 \end{aligned}$$